

ED-FEED: An AI-Driven Intelligent Student Feedback Analysis and Insight Generation System

Baseline Model Evaluation

The evaluation of the baseline sentiment analysis model focused on improving data representation, preprocessing quality, and target label consistency. The model was built using TF-IDF vectorization with a limit of 5,000 features and bigrams (1,2), enabling it to capture both individual words and contextual word pairs for better sentiment understanding.

A key part of the evaluation was feature engineering, where two preprocessing approaches were analyzed. The original cleaning method removed all non-alphabetic characters, which resulted in the loss of important sentiment indicators such as punctuation. To address this, an improved preprocessing technique was implemented that retained symbols like ‘!’ and ‘?’, helping preserve emotional intensity in the text. This enhancement contributed to better sentiment interpretation, especially in strongly opinionated reviews.

Another important step was label mapping, where numerical star ratings were converted into categorical sentiment labels: positive (4-5), neutral (3), and negative (1-2). This ensured that the model could be trained effectively in a supervised learning setup with clearly defined target classes.

The model was trained using Logistic Regression, and its performance was evaluated using classification metrics. The results showed an overall accuracy of approximately 82%, indicating good general performance. The model demonstrated strong performance in identifying positive sentiment and reasonably good performance for negative sentiment. However, it showed poor performance in detecting neutral sentiment, with very low recall. This limitation is primarily due to class imbalance, as the dataset contains significantly fewer neutral samples compared to positive and negative ones.

The entire pipeline—from text preprocessing to feature extraction and prediction—was thoroughly validated to ensure logical consistency and correctness. Edge case testing further confirmed that the model relies on textual sentiment rather than solely on ratings, improving its real-world applicability.

In conclusion, the baseline model is functionally reliable and performs well overall, particularly for positive and negative sentiment classification. The improvements in preprocessing enhanced the model’s ability to capture sentiment effectively. However, further optimization is required to address class imbalance and improve neutral sentiment detection.

RESULT

Sentiment Label Distribution

Sentiment	Count
Positive	11804
Negative	6784
Neutral	1389

The dataset is imbalanced, with a higher number of positive samples compared to negative and significantly fewer neutral samples.

Classification Report

Class	Precision	Recall	F1-Score	Support
Negative	0.80	0.76	0.78	6784
Neutral	0.77	0.04	0.08	1389
Positive	0.83	0.95	0.88	11801

Overall Performance

Metric	Value
Accuracy	0.82
Macro Avg F1	0.58
Weighted Avg F1	0.79

The model achieved an overall accuracy of 82%, indicating good performance in predicting sentiment on the dataset.